

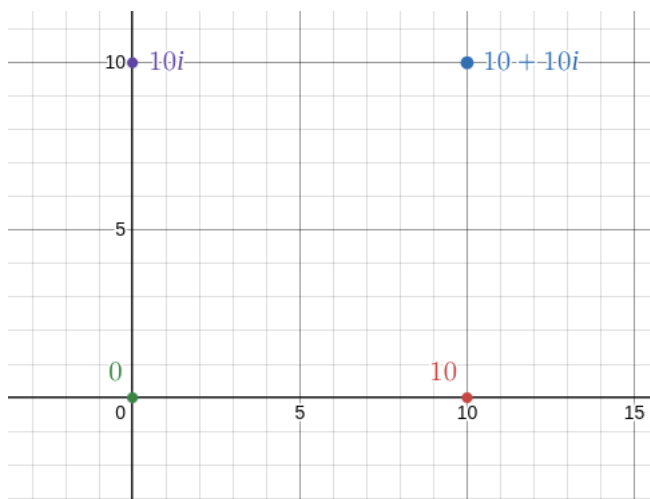
Journal for Complex Analysis

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September 23, 2026

Chapter 1

Task 10 Find four complex numbers such that when placed on an Argand diagram, they are located at the vertices of a square.



Here the points I picked were:
 $10, 10 + 10i, 10i, 0$.

Task 11 Show that the expression for $z + w$ and $z * w$ in the previous definition match the results you get by treating each complex number as a binomial and adding or multiplying them according to the rules of polynomials, replacing i^2 with -1 wherever it appears.

There are two parts to this question: $z * w$ and $z + w$

$$z = x + yi \text{ and } w = u + vi$$

Starting with $z * w$:

$$z * w = (x + yi) * (u + vi)$$

$$z * w = x * u + u * yi + x * vi + y * v * i^2$$

$$z * w = x * u + u * yi + x * vi - y * v$$

$$z * w = (x * u - y * v) + (u * y + x * v)i$$

Next $z + w$:

$$z + w = (x + yi) + (u + vi)$$

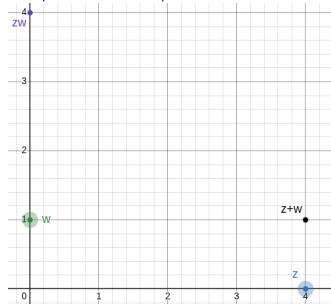
$$z + w = x + u + yi + vi$$

$$z + w = (x + u) + (y + v)i$$

Task 12 For each pair of complex numbers z and w given below, compute $z + w$ and zw . Then plot all four numbers $z, w, z + w$ and zw on an Argand diagram.

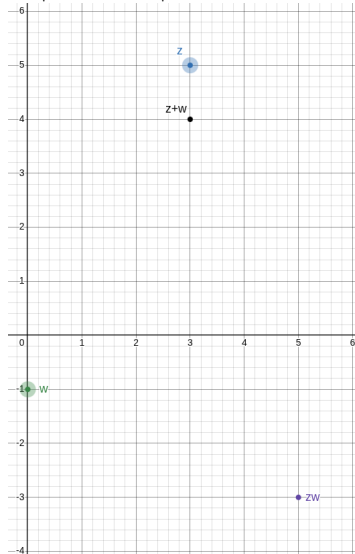
a $z = 4, w = i$

$z + w = 4 + i$ and $z * w = 4i$



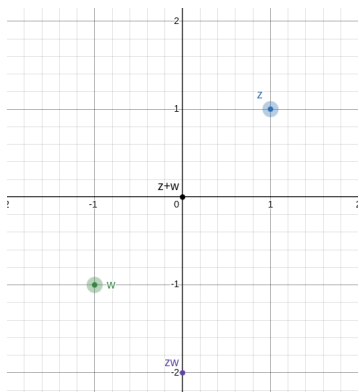
b $z = 3 + 5i, w = -i$

$z + w = 3 + 4i$ and $z * w = -5 - 3i$

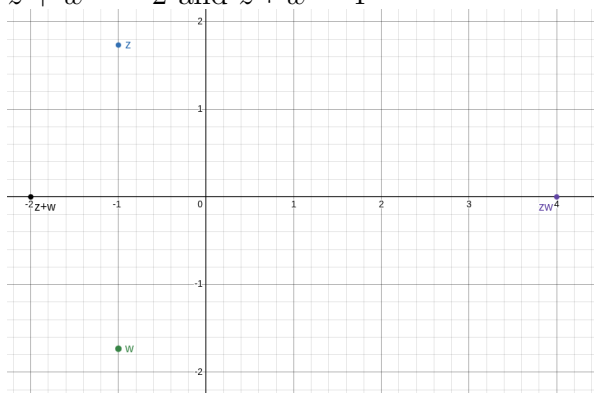


c $z = 1 + i, w = -1 - i$

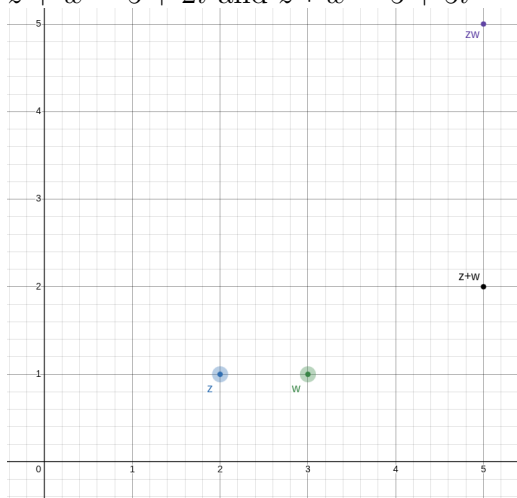
$z + w = 0$ and $z * w = -2i$



d $z = -1 + i\sqrt{3}, w = -1 - i\sqrt{3}$
 $z + w = -2$ and $z * w = 4$



e $z = 2 + i, w = 3 + i$
 $z + w = 5 + 2i$ and $z * w = 5 + 5i$



Task 13 Suppose $z, w \in \mathbb{C}$. Show that $z + w = w + z$ and $z * w = w * z$.

There are two proofs within this question: $z + w = w + z$ and $z * w = w * z$.
 $w = a + bi$

$$z = c + di$$

$$z + w = w + z$$

$$a + bi + c + di = c + di + a + bi.$$

Since there is only real numbers contained within a, b, c, d , the rules of addition will apply.

$$z * w = w * z$$

$$(a + bi)(c + di) = (a * c + a * di + c * bi - b * d)$$

Since there is only real numbers contained within a, b, c, d , the rules of multiplication will apply.

Task 14 Suppose that $z_1, z_2, w \in \mathbb{C}$. Show that $w(z_1 + z_2) = wz_1 + wz_2$.

$$w = a + bi$$

$$z_1 = c + di$$

$$z_2 = e + fi$$

$$w * (z_1 + z_2) = (a + bi)(c + di + e + fi)$$

$$w * (z_1 + z_2) = a * c + a * e + a * di + a * fi + c * bi - b * d + e * bi - b * f$$

$$w * (z_1 + z_2) = (a * c - b * d + c * bi + a * di) + (a * e - b * f + e * bi + a * fi)$$

$$w * z_1 + w * z_2 = (a + bi)(c + di) + (a + bi)(e + fi)$$

$$w * z_1 + w * z_2 = (a * c + a * di + c * bi - b * d) + (a * e + a * fi + e * bi - b * f)$$

$$w * z_1 + w * z_2 = (a * c + a * di + c * bi - b * d) + (a * e + a * fi + e * bi - b * f) = w * (z_1 + z_2)$$

Thus, $w * z_1 + w * z_2 = w * (z_1 + z_2)$.

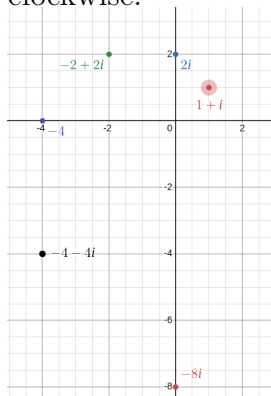
Task 15

$$(1 + i)^2 = 2i$$

$$(1 + i)^3 = -2 + 2i$$

$$(1 + i)^4 = -4$$

When plotting the points on a graph, the graph produced a spiral type shape that was counter clockwise.



The same happened when plotting the $(1 - i\sqrt{3})$, the only difference was the spiral that was

produced went clockwise instead of counter clockwise.



Task 16

The basic pattern I noticed was that for the first four powers of i , they were unique values. However, after that, the values tend to repeat in a predictable pattern.

$$\begin{aligned} i^{99} &= -i \\ i^{1001} &= i \\ i^{57368} &= 1 \end{aligned}$$

Task 17

$$\begin{aligned} z * z^{-1} &= (x + yi)\left(\frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i\right) \\ z * z^{-1} &= \frac{x^2}{x^2+y^2} - \frac{x*yi}{x^2+y^2} + \frac{x*yi}{x^2+y^2} + \frac{y^2}{x^2+y^2} \\ z * z^{-1} &= \frac{x^2+y^2}{x^2+y^2} \\ z * z^{-1} &= 1 \end{aligned}$$

Task 18

$$\begin{aligned} i^{-1} &= -i \\ (3 + 4i)^{-1} &= \frac{3}{25} - \frac{16}{25}i \\ \left(\frac{1}{2} - \frac{1}{2}i\right)^{-1} &= 1 + 1i \\ \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^{-1} &= \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) \end{aligned}$$

Task 19

Task 11, 13, 14, and 17 all prove rules F1, F2, F3, and F6. As well as a 1 and a 0 being included within this set, that proves F4. We also know that a or b must be a real number, which has a additive inverse, which proves F5.

Thus, $\mathbb{C}$ is a field.

Task 20

a

$$4z = -8i$$

$$z = -2i$$

b

$$4iz = -8i$$

$$z = -2$$

c

$$4iz = -8$$

$$z = -2i \exp(-1) = 2i$$

d

$$z(1 - i) = 4$$

$$z = \frac{4}{1-i}$$

e

$$z^2 = -4$$

$$z = 2i$$

Task 21

Lets assume that the real component of the z is used to measure the relation between two different z objects. Then it would follow that the bigger the real value on one object would mean that the object comes after the one with the smaller real value.

However, OF2 would break this continuity.

Given an object with a large imaginary component and an even real number, when multiplied with another imaginary number, the result would be a negative number which would break OF2.

Thus, $\mathbb{C}$ is not an ordered field.

Task 22

a

$$5 - 3i$$

b

$$-5 + 3i$$

c

$$3i(1 - 2i) = 6 + 3i$$

d

$$-3i(1 + 2i) = 6 - 3i$$

e

$$-i^{99} = -i^3 = i$$

Task 23

$$z = x + yi$$

$$\bar{z} = x - yi$$

$$w = a + bi$$

$$\bar{w} = a - bi$$

a

$$\overline{x - yi} = x + yi$$

Here the conjugate of the conjugate is the original object.

b

$$x - yi + a - bi = \overline{x + yi} + \overline{a + bi}$$

Here the addition of the conjugates is the same as taking the conjugates of the objects individually.

c

$$w * z = a * x + a * yi + x * bi - y * b$$

$$\overline{w * z} = a * x - y * b - a * yi - x * bi$$

$$\bar{w} * \bar{z} = (x - yi)(a - bi)$$

$$\bar{w} * \bar{z} = a * x - x * bi - a * yi - y * b$$

$$\bar{w} * \bar{z} = a * x - x * bi - a * yi - y * b = \overline{w * z}$$

Here, both sides of the equation produce the same result when expanded.

d

$$z + \bar{z} = x + yi + x - yi$$

$$z + \bar{z} = 2x$$

$$2\Re(z) = 2(x)$$

$$2\Re(z) = 2x = z + \bar{z}.$$

Here, both sides of the equation produce the same result when expanded.

e

$$z - \bar{z} = x + yi - (x - yi)$$

$$z - \bar{z} = yi + yi$$

$$z - \bar{z} = 2yi$$

$$2i\Im(z) = 2i(y)$$

$$2i\Im(z) = 2iy = z - \bar{z}$$

Here, both sides of the equation produce the same result when expanded.

f

$$z^{-1} = \frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i$$

$$\overline{z^{-1}} = \frac{x}{x^2+y^2} + \frac{y}{x^2+y^2}i$$

$$\bar{z}^{-1} = \frac{x}{x^2+(-y)^2} - \frac{-y}{x^2+(-y)^2}$$

$$\bar{z}^{-1} = \frac{x}{x^2+(-y)^2} + \frac{y}{x^2+(-y)^2} = \overline{z^{-1}}$$

Here, both sides of the equation produce the same result when expanded.

Task 24

Describe the geometric shape defined by the given equations.

$$z = x + yi$$

$$\bar{z} = x - yi$$

Within a standard graph, we represent the x and the y as coordinates, we can do the same with the complex, where x equals x and yi equals y in the 2D graph space.

a

$$x + yi = x - yi$$

$$x = x$$

This functionally represents a 1D line, where only x values live.

b

$$x + yi = i(x - yi)$$

$$x + yi = y + xi$$

Here, the variables equal one another, similar to $y = x$ in the 2D graph space.

c

$$x + yi + x - yi = 1$$

$$2x = 1$$

$$x = \frac{1}{2}$$

There is only one point and that is $\frac{1}{2}$
d

$$(x + yi)^2 + (x - yi)^2 = 2$$

$$x^2 + 2x * yi + y^2 + x^2 - 2x * yi + y^2 = 2$$

$$x^2 + y^2 + x^2 + y^2 = 2$$

$$2 * x^2 + 2 * y^2 = 2$$

This is a standard unit circle.

e

$$(x + yi)^{-1} = x - yi$$

$$1 = (x - yi) * (x + yi)$$

$$1 = x^2 - y^2$$

This is a hyperbolic function on the plane.

Task 25

$$z = x + yi$$

$$\bar{z} = x - yi$$

$$z * \bar{z} = (x + yi)(x - yi)$$

$$z * \bar{z} = x^2 - x * yi + x * yi - y^2 i^2$$

$$z * \bar{z} = x^2 + y^2$$

In this case, $\sqrt{z * \bar{z}}$ would be the distance from the origin, giving the total length of the vector.

Task 26

a

$$|z| = \sqrt{i * -i}$$

$$|z| = 1$$

b

$$|z| = \sqrt{1^2 - i^2}$$

$$|z| = \sqrt{2}$$

c

$$|z| = \sqrt{(-3)^2}$$

$$|z| = 3$$

d

$$|z| = \sqrt{9+3}$$

$$|z| = \sqrt{12}$$

e

$$|z| = \sqrt{25+144}$$

$$|z| = \sqrt{169}$$

$$|z| = 13$$

Task 27

$$z = x + yi$$

$$\bar{z} = x - yi$$

$$|z| = \sqrt{x^2 + y^2}$$

$$z^{-1} = \frac{\bar{z}}{|z|^2}$$

$$\frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i = \frac{x-yi}{x^2+y^2}$$

Task 28

$$z = x + yi$$

$$|z| = \sqrt{x^2 + y^2}$$

$$w = a + bi$$

$$|w| = \sqrt{a^2 + b^2}$$

a $|Re z| \leq |z|$

$$x \leq \sqrt{x^2 + y^2}$$

This always going to be true.

b $|Im z| \leq |z|$

$$y \leq \sqrt{x^2 + y^2}$$

Similiar to the last problem, this is always going to be true.

c $Re(z\bar{w}) \leq |z||w|$

$$(x+a) \leq \sqrt{(x)^2 + (y)^2} * \sqrt{(a)^2 + (b)^2}$$

Similiar to the last problem, this is always going to be true.

d $|z + w| \leq |z| + |w|$

$$|x + yi + a + bi| \leq \sqrt{(x^2 + y^2) + (a^2 + b^2)}$$

$$\sqrt{(x+a)^2 + (y+b)^2} \leq \sqrt{(x^2 + y^2)} + \sqrt{(a^2 + b^2)}$$

Similiar to the last problem, this is always going to be true.

Task 29 Prove the triangle inequality: if $a, b, c \in \mathbb{C}$

$$|a - c| \leq |a - b| + |b - c|$$

$$|a + (b - b) - c| \leq |a - b| + |b - c|$$

$$|a - b + b - c| \leq |a - b| + |b - c|$$

Here we see that the inequality must be true because we can add b within.

Task 30 Show that if $z_1, z_2, \dots, z_n$ is any finite collection of complex numbers, then

$$|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|.$$

$$z_n = a_n + ib_n$$

Starting with two different z values first we get, $\sqrt{(a_1 + a_2)^2 + (b_1 + b_2)^2} \leq \sqrt{a_1^2 + b_1^2} + \sqrt{a_2^2 + b_2^2}$.

Let, $z_1 = 2 + 2i$ and $z_2 = 2 - 4i$.

Substituting the values in we get, $\sqrt{(2+2)^2 + (2-4)^2} \leq \sqrt{2^2 + (2)^2} + \sqrt{2^2 + (-4)^2}$.
 $\sqrt{16 + 4} \leq \sqrt{8} + \sqrt{20}$.

Here it is shown that the inequality has held for this example.

Here, we know that this must hold because when we add to imaginary vectors, they could point in different directions of each other. This would cause the shortest distance between 0 and the combined vectors to be shorter than the combined vectors.

Adding more vectors would only change the shortest path of the combine vectors.

Thus, the inequality must hold for a finite collection of complex numbers.

Task 31

$$z = x + yi$$

$$|z| = \sqrt{x^2 + y^2}$$

$$w = a + bi$$

$$|w| = \sqrt{a^2 + b^2}$$

$$\begin{aligned} |z * w| &= |x * a + a * yi + x * bi - y * b| \\ |z * w| &= \sqrt{x^2 * a^2 + y^2 * b^2 + a^2 * y^2 + x^2 * b^2} \end{aligned}$$

$$\begin{aligned} |z| * |w| &= |x + yi| * |a + bi| \\ |z| * |w| &= \sqrt{x^2 + y^2} * \sqrt{a^2 + b^2} \\ |z| * |w| &= \sqrt{x^2 + y^2} * \sqrt{a^2 + b^2} \\ |z| * |w| &= \sqrt{a^2 * x^2 + b^2 * y^2 + y^2 * a^2 + x^2 * b^2} \\ |z| * |w| &= \sqrt{a^2 * x^2 + b^2 * y^2 + y^2 * a^2 + x^2 * b^2} = |z * w| \end{aligned}$$

Task 32 Suppose $z, w \in \mathbb{C}$. Let T_1 be the triangle with vertices 0, 1, and z and let T_2 be the triangle with vertices 0, w , and zw . Show that T_1 and T_2 are similar triangles.

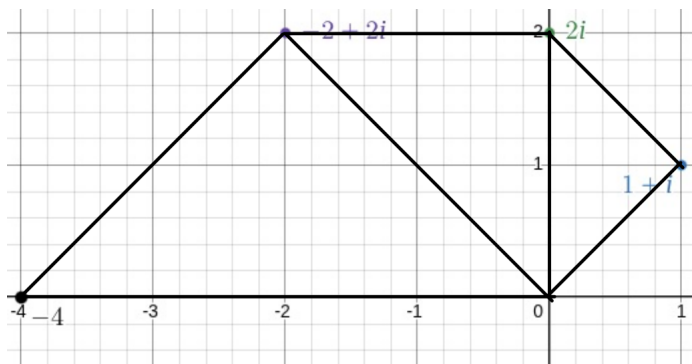
$$z = x + yi, w = a + bi$$

For this I'm going to use Side-Side-Side similarity. To prove this we have to prove that the sides of the triangle are scaled by a factor of some scalar. In this case, this provided to us with the complex w number.

We could describe the vertices of w and zw and divide off the w and show that the only possible answer would have to be T_1 triangle.

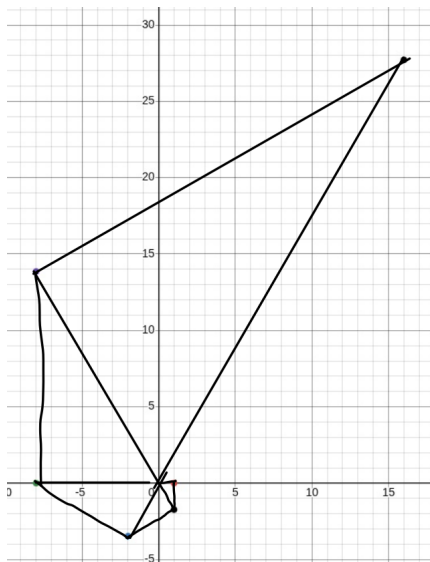
Thus, the triangles are similar.

Task 33 Using the triangle visualization of multiplication from task 32, explain how the figure below relates to the sequence $(1 + i)^n, 1 \leq n \leq 4$. Draw the corresponding picture for $(1 - i\sqrt{3}), 1 \leq n \leq 4$.



This is for the sequence of $(1 + i)^n$

This picture seems to be neatly expanding, the angle between the triangles are 45 degrees apart.



This is for the sequence of $(1 - \sqrt{3}i)^n$

Here, the picture seems to spiral out, and within task 15, this made sense because the numbers

seemed to explode outwards.

Task 34 Describe what geometric shape is defined by each of these equations.

$$z = x + yi$$

a $|z| = 1$

This is a unit circle, we can describe this as the magnitude equal 1 at any given x and y , which would make a circle.

b $|z - 1| = 2$

This is a circle with the radius 2 and shifted 1 to the right, or centered on $(1, 0)$.

c $|z + 4i| = 3$

This is a circle with a radius of 3 shifted down 4. Centered on $(0, -4)$.

d $|z| = |z - 1|$

This is a line on $x = 0.5$.

e $|z - i| = |z + i|$

This is a line on $y = 0$.

f $|z - 2| = \operatorname{Re} z$

This is a sideways opening parabola, opened to the right, where the vertex is $(1, 0)$.

Task 35 Show that if z is a nonzero complex number, then $z/|z|$ has modulus 1. Describe geometrically how $z/|z|$ relates to z .

$$z = x + yi$$

$$|z| = \sqrt{x^2 + y^2}$$

Before solving the question algebraically, we can discuss what this equation means.

If we think about z being a vector of sorts, then $z/|z|$ would be the taking the vector and dividing off the length of the vector. Since both vectors have the same length, they are 1 length relative to each other.

So geometrically this would mean taking your complex number and scaling it to the length of itself.

$$\left| \frac{z}{|z|} \right| = \frac{x+yi}{\sqrt{x^2+y^2}}$$

$$\left| \frac{z}{|z|} \right| = x|z|^{-1} + yi|z|^{-1}$$

$$\left| \frac{z}{|z|} \right| = \sqrt{x^2|z|^{-2} + y^2|z|^{-2}}$$

$$\left| \frac{z}{|z|} \right| = \sqrt{\frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2}}$$

$$\left| \frac{z}{|z|} \right| = \sqrt{\frac{x^2+y^2}{x^2+y^2}}$$

$$\left| \frac{z}{|z|} \right| = \sqrt{1} = 1$$

Task 36 Show that, if θ is any real number, then $\cos \theta + i \sin \theta$ has modulus 1.

Since, $\cos$ and $\sin$ both correlate to the unit circle, what ever the theta value is, together cosine and sine will add up to be the radius of the circle.

$$\begin{aligned} |\cos \theta + i \sin \theta| &= (\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta) \\ |\cos \theta + i \sin \theta| &= (\cos \theta)^2 + (\sin \theta)^2 \\ |\cos \theta + i \sin \theta| &= 1 \end{aligned}$$

Task 37 Find the principal argument and one other argument for each number in task 26.

a $z = i$

Since this is a simple rotation on the imaginary axis, $\text{Arg } z = \frac{\pi}{2}$.

Another $\text{Arg } z$ is $\frac{5\pi}{2}$.

b $z = 1 + i$

Breaking this up into a triangle, it is easy to see that the triangle will have a base of 1 and height of i . If we then scale the triangle to the unit circle, we can see that this new triangle has $\frac{1}{\sqrt{2}}$ and $\frac{i}{\sqrt{2}}$. Or, we could see that the hypotenuse is $\sqrt{2}$, which only happens on a 45 degree or $\frac{\pi}{4}$ triangle.

So, $\text{Arg } z = \frac{\pi}{4}$ and $\frac{9\pi}{4}$.

c $z = -3$

Since this is a simple rotation on the real axis, $\text{Arg } z = \pi$.

Another $\text{Arg } z$ is 3π .

d $z = 3 - i\sqrt{3}$

Taking the modulus on this to find the hypotenuse, shows that this triangle is 30 60 90 triangle. This is information, $\text{Arg } z = \frac{\pi}{6}$ and $\frac{13\pi}{6}$.

e $z = -5 + 12i$

Taking the modulus shows that there is no easy pattern here, the given triangle is a pythagorean triple. However, we are given the lengths of the sides, so we can use sine or cosine for the angle. I chose sine and did $\sin^{-1} -5/13 + pi$, which equals -0.384615384615 . Adding two π would result in 5.89856992256

Task 38 Write each of the numbers from task 26 in polar form.

Using the last task, we have the θ for each set of numbers.

a $z = i$

$$1(\cos(\pi) + i \sin(\pi))$$

b $z = 1 + i$

$$\sqrt{2}(\cos(\frac{\pi}{4}) + i \sin(\frac{\pi}{4}))$$

$$\mathbf{c} \ z = -3$$

$$3(\cos(\pi) + i \sin(\pi))$$

$$\mathbf{d} \ z = 3 - i\sqrt{3}$$

$$2\sqrt{3}(\cos(\frac{\pi}{6}) + i \sin(\frac{\pi}{6}))$$

$$\mathbf{e} \ z = -5 + 12i$$

$$13(\cos(-0.384615384615) + i \sin(-0.384615384615))$$

Task 39 Find general formulas that convert between the rectangular form $x + yi$ and the polar $r(\cos \theta + i \sin \theta)$ of a complex number, and explain why they work.

For the formulas, $r = |z|$ was already given within the definition 1.4.2.

However, $\text{Arg } z$ was not provided a formula, but we can use sine and cosine to solve for the angle between the triangle. Since we can describe the real and imaginary parts as instructions for constructing the sides of a triangle, each side length corresponds with the real or imaginary part. Using this knowledge we can find hypotenuse and complete the triangle. Then use either sine, cosine, or tangent to find the angle.

$$\text{So, } \frac{\Re z}{|z|} = \cos \theta$$

$$\cos^{-1} \frac{\Re z}{|z|} = \theta$$

Taking both cosine and sine would be very useful here because there will sometimes in which one angle will not describe both x and y .

Task 40 Use the angle sum formulas to show that for all $\alpha, \beta \in \mathbb{R}$, then

$$(\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) = \cos(\alpha + \beta) + i \sin(\alpha + \beta).$$

Angle Sum Formulas:

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta,$$

$$\sin(\alpha + \beta) = \cos \alpha \sin \beta + \sin \alpha \cos \beta.$$

This is pretty much given to the reader, by just looking at the formulas, we can see that the right side of the equation will convert to the first sum angle formulas.

$$(\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) = \cos(\alpha + \beta) + i \sin(\alpha + \beta).$$

$$\cos(\alpha) \cos(\beta) + i \cos(\alpha) \sin(\beta) + i \sin(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) = \cos(\alpha + \beta) + i \sin(\alpha + \beta).$$

$$\cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) + i(\cos(\alpha) \sin(\beta) + \sin(\alpha) \cos(\beta)) = \cos(\alpha + \beta) + i \sin(\alpha + \beta).$$

Here this can be reduced using the formulas.

$$\cos(\alpha + \beta) + i(\sin(\alpha + \beta)) = \cos(\alpha + \beta) + i \sin(\alpha + \beta).$$

Thus this statement is true.

Task 41

The formula could be derived from finding both the z and w complex number polar forms and realising that the answer to the idea of similar triangles is that zw must be equal to the polar form multiplication.

Task 42 Show that *de Moivre's formula* is true: if $\theta \in \mathbb{R}$ and $n \in \mathbb{N}$, then
 $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$.

Using task 39, we can see that doing the multiplication out, we would end up with a similar set of circumstances. We could say that θ could be both α and β , and that we could just represent every power as a multiplication of two different angles. Since we add both angles, this would lead to $n\theta$.

$$\begin{aligned} (\cos \theta + i \sin \theta)(\cos \theta + i \sin \theta)(\cos \theta + i \sin \theta)^{n-2} &= \cos(n\theta) + i \sin(n\theta) \\ (\cos(\theta) \cos(\theta) - \sin(\theta) \sin(\theta) + i(\cos(\theta) \sin(\theta) + \sin(\theta) \cos(\theta))) &(\cos \theta + i \sin \theta)^{n-2} = \\ \cos(n\theta) + i \sin(n\theta) \\ (\cos(2\theta) + i \sin(2\theta))(\cos \theta + i \sin \theta)^{n-2} &= \cos(n\theta) + i \sin(n\theta) \end{aligned}$$

And the process can be repeated to n amounts.

Task 43 Find $(1+i)^n$ and $(1-i\sqrt{3})$ for $n = 10, 25, 99$. Write your answers in the form $x + yi$.

$$(1+i)^{10} = 32(\cos(\frac{\pi}{2}) + i \sin(\frac{\pi}{2}))$$

Here I simplified the fractional part of cosine and sine.

$$(1+i)^{10} = 32((0) + i(1))$$

$$(1+i)^{25} = \sqrt{2}^{25}(\cos(\frac{\pi}{4}) + i \sin(\frac{\pi}{4}))$$

$$(1+i)^{25} = \sqrt{2}^{25}(\sqrt{2}) + i(\sqrt{2})$$

$$(1+i)^{25} = \sqrt{2}^{24} + i\sqrt{2}^{24}$$

$$(1+i)^{25} = 4096 + 4096i$$

$$(1+i)^{99} = \sqrt{2}^{99}(\cos(\frac{3\pi}{4}) + i \sin(\frac{3\pi}{4}))$$

$$(1+i)^{99} = \sqrt{2}^{99}((-\frac{1}{\sqrt{2}}) + i(\frac{1}{\sqrt{2}}))$$

$$(1+i)^{99} = -\sqrt{2}^{98} + i\sqrt{2}^{98}$$

$$(1+i)^{99} = -2^{49} + 2^{49}i$$

$$(1-i\sqrt{3})^{10} = 2^{10}(\cos(\frac{2\pi}{3}) + i \sin(\frac{2\pi}{3}))$$

$$(1-i\sqrt{3})^{10} = 2^{10}(-\frac{1}{2}) + i\frac{\sqrt{3}}{2}$$

$$(1-i\sqrt{3})^{10} = -2^9 + i2^9\sqrt{3}$$

$$(1-i\sqrt{3})^{25} = 2^{25}(\cos(\frac{\pi}{3}) + i \sin(\frac{\pi}{3}))$$

$$(1-i\sqrt{3})^{25} = 2^{25}(\cos(\frac{\pi}{3}) + i \sin(\frac{\pi}{3}))$$

$$(1-i\sqrt{3})^{25} = 2^{25}(-\frac{1}{2}) + i(\frac{\sqrt{3}}{2})$$

$$(1 - i\sqrt{3})^{25} = -2^{24} + 2^{24}i\sqrt{3}$$

$$(1 - i\sqrt{3})^{99} = 2^{99}(\cos(\frac{99\pi}{3}) + i\sin(\frac{99\pi}{3}))$$

$$(1 - i\sqrt{3})^{99} = 2^{99}(\cos(33\pi) + i\sin(33\pi))$$

$$(1 - i\sqrt{3})^{99} = 2^{99}(-1)$$

$$(1 - i\sqrt{3})^{99} = -2^{99}$$